Tim Richards

Senior Lecturer II

(508) 208-0301

richards@cs.umass.edu

http://www.cs.umass.edu/~richards

Manning College of Information and Computer Sciences
Department of Computer Science
University of Massachusetts Amherst
Amherst, MA 01003-9264

Research

My research interests center on questions related to operating systems, programming language implementation, the web, scalability, sustainability, and education. I have a broad interest in computing at all levels across all disciplines.

Education

Ph.D. in Computer Science, University of Massachusetts Amherst, February 2010
Thesis: *Generalized Instruction Selection Generation: The Automatic Construction of Instruction Selectors from Descriptions of Compiler Internal Forms and Target Machines.* Advisors: J. Eliot B. Moss and Charles C. Weems

M.S. in Computer Science, University of Massachusetts Amherst, Spring 2006
Master's Thesis: *Proving the Equivalence of Instruction Semantics*
Advisors: J. Eliot B. Moss and Charles C. Weems

B.A. in Computer Science, Magna Cum Laude, Clark University, 1999

Professional Experience

- **Senior Lecturer II**, University of Massachusetts Amherst, Amherst, MA, 2021 – today
- **Senior Lecturer I**, University of Massachusetts Amherst, Amherst, MA, 2017 – 2021
- **Lecturer**, University of Massachusetts Amherst, Amherst, MA, 2011 – 2017
- **Visiting Assistant Professor**, Trinity College, Hartford, CT, 2009 – 2011
- **Adjunct Faculty Member**, Clark University, Worcester, MA, 2007 – 2008
- **Adjunct Faculty Member**, Springfield College, Springfield, MA, 2007 – 2008
- **Graduate Research Assistant**, University of Massachusetts Amherst, 2002 – 2009
- **Software Engineer II**, Concord Communications, Marlboro, MA, 2000 – 2001
- **Systems Software Engineer**, Digital Equipment Corporation, Shrewsbury, MA, 1999 – 2000

Achievements and Honors

Distinguished Teaching Award Nominee 2022, University of Massachusetts Amherst
Flexible Learning Fellow 2021-2022
Manning Prize Nomination 2020
Professional Improvement Leave Award 2019 (t.ly/YRp83) p. 2
Innovation Fellow 2018 (t.ly/2wbY9)
College Outstanding Teaching Award (COTA) 2018 (t.ly/wmVAN)
Distinguished Teaching Award Nominee 2018, University of Massachusetts Amherst
Distinguished Teaching Award Nominee 2017, University of Massachusetts Amherst
Thomas Jefferson Award Recipient, University Park Campus School, Worcester Public Schools

Teaching

University of Massachusetts Amherst, Manning College of Information and Computer Sciences

COMPSCI 198C Introduction to the C Prog. Language F20, W21, S21, Su21, F21, W22, S22, Su22, F22, W23

COMPSCI 187 Programming with Data Structures	Su12, Su13, S14, Su14, Su15, Su16, Su17, Su18, Su19, Su20
COMPSCI 220 Programming Methodology	F12, F14, S15, F15
COMPSCI 230 Computer Systems Principles	F11, S12, F12, S13, F13, S14, F14, S15, F16, S17, S18, S19, S20, S21
COMPSCI 326 Web Programming	S12, Su12, F12, S13, F13, S14, F14, S15, F15, S16, F16, S17, F17, S18, F18, S19, F19, S22, Su22, S23
COMPSCI 377 Operating Systems	S16, F17, F18, F19, F20, S21, S22, Su22, F22, S23
COMPSCI 445 Information Systems	F11
COMPSCI 497 Scalable Web Systems	S15, F15, S16, F19, F20, S21, F22
COMPSCI 591A Agile Software Project Management	F16

Trinity College, Department of Computer Science

CPSC 110 Computers, Information, & Society	F09, S10, S11
CPSC 225 Client-side Web Programming	S11
CPSC 225 Server-side Web Programming	F10
CPSC 333 Computer Networks	S10
CPSC 371 Compiler Techniques	S11
CPSC 372 Database Fundamentals	F09, F10
CPSC 415 Advanced Web Programming	S10

Clark University, Department of Math and Computer Science

CSCI 230 Compiler Design	S08

Department of Mathematics, Physics, & Computer Science, Springfield College

CISC 105 Introduction to Computers	F07

Publications

Books

1. Nell Dale, Chip Weems, and Tim Richards, Programming and Problem Solving with C++ 7th Edition, Jones & Bartlett Learning, 2022, 1273 pp. ISBN-13 978-1284157321
2. Nell Dale, Chip Weems, and Tim Richards, *C++ Plus Data Structures 6th Edition*, Jones & Bartlett Learning, 2016, 838 pp. ISBN 978-1284089189.

Conferences

1. Paul E. Dickson, Tim Richards, and Brett A. Becker. 2022. Experiences Implementing and Utilizing a Notional Machine in the Classroom. In Proceedings of the 53rd ACM Technical Symposium on Computer Science Education V. 1 (SIGCSE 2022). Association for Computing Machinery, New York, NY, USA, 850–856. https://doi.org/10.1145/3478431.3499320
2. Keith Chapman, Tony Hosking, Eliot Moss, and Tim Richards, "Closed and Open Nested Atomic Actions for Java: Language Design and Prototype Implementation", in the 2014 International Conference on Principles and Practice of Programming on the Java Platform: Virtual Machines, Languages, and Tools (PPPJ), pp. 169-180, September 2014. (39% acceptance rate)

Journals

1. CASL: A Rapid-prototyping Language for Modern Micro-Architectures, Edward K. Walters II, J. Eliot B. Moss, Trek Palmer, Timothy Richards, and Charles C. Weems. In Computer Languages, Systems, and Structures, 2007, Volume 34/4, pp. 195-211
2. CISL: A Class-based Machine Description Language for Co-Generation of Compilers and Simulators, J. Eliot B. Moss, Trek Palmer, Timothy Richards, Edward K. Walters II, and Charles C. Weems, In the

International Journal of Parallel Programming, Springer-Verlag, Volume 33, Numbers 2-3, pp. 231-246, June 2005.

Workshops

1. Improved Whiteboard Processing for Lecture Capture, Paul E. Dickson, Chris Kondrat, W. Richards Adrion, Tim Richards, and Ryan B. Szeto, in the 11th International Workshop on Multimedia Technologies for E-Learning (MTEL), San Jose, CA, December 2016.
2. Portable Lecture Capture that Captures the Complete Lecture, Ryan Szeto, Rick Adrion, Tung Pham, Tim Richards, Paul Dickson, and Chris Kondrat, in the 10th IEEE International Workshop on Multimedia Technologies for E-Learning (MTEL), Miami, FL, December 2015.
3. Thinking in Parallel: Hardware to Software - Adopting the TCPP Core Curriculum in Computer Systems Principles, Timothy Richards, in the Third NSF/TCPP Workshop on Parallel and Distributed Computing Education (EduPar-13), Boston, MA, May 2013.
4. The CoGenT Project: Co-Generating Compilers and Simulators for Dynamically Compiled Languages, J. Eliot B. Moss, Charles C. Weems, and Timothy Richards. In Proceedings of the 2003 International Parallel and Distributed Processing Symposium Workshop on Next Generation Software, Nice, France, April 2003.
5. Towards Universal Code Generator Generation, Timothy Richards, Edward K. Walters II, J. Eliot B. Moss, Trek Palmer, and Charles C. Weems. In Proceedings of the 2008 International Parallel and Distributed Processing Workshop on Next Generation Software, Miami, FL, April 2008.
6. CMDL: A Class-based Machine Description Language for Co-Generation of Compilers and Simulators, J. Eliot B. Moss, Trek Palmer, Timothy Richards, Edward K. Walters II, and Charles C. Weems. In Proceedings of the 2004 International Parallel and Distributed Processing Symposium Workshop on Next Generation Software, Santa Fe, NM, April 2004.

Technical Reports

1. A Unified Framework for the Automatic Generation of System Tools and Components, Timothy Richards, Edward K. Walters II, Trek Palmer, J. Eliot B. Moss, and Charles C. Weems. UMass CS Technical Report TR-07-56, 2007.

Posters and Panels

1. Innovation with Scale: Turning Large Class Sizes into Opportunities for Pedagogical Innovation, Neena Thota, Tim Richards, Marc Liberatore, Gordon Anderson, William T. Verts, and Michelle Trim, in the Consortium for Computing Sciences in Colleges-Northeastern (CCSC), Panel Session, 2017.
2. PAOL and Lecture-Viewer, Brendan Murphy, Tung Pham, Raelen Recto, Ryan Szeto, Rick Adrion, Timothy Richards, and Paul Dickson, in the Conference on the American Society for Engineering Education, Poster Session, 2014.

Funding and Support

1. CICStart: Flexible Learning in Introductory Computing Courses, Flexible Learning Fellows Program, University of Massachusetts, $30,000, co-PI with Joe Chiu, 2021-2022.
2. Micro Virtual Machines for Managed Languages, National Science Foundation CSR Medium, $170,000, co-PI with Eliot Moss of UMass Amherst and Tony Hosking of Purdue, 2014-2017.
3. Auto-Grading with Useful Student Feedback, Flex Grant for Teaching/Faculty Development University of Massachusetts Amherst, $500, Tim Richards, 2015.
4. Geospatial Programming: Child's Play, Google Unrestricted Award, $18,000, co-PI with Dr. Dana Tomlin of University of Pennsylvania School of Design and Yale School of Forestry and Environmental Studies, 2013.

5. Portable Performance for Parallel Managed Languages, National Science Foundation CSR Medium, $800,000, co-PI with Eliot Moss of UMass Amherst and Tony Hosking of Purdue, 2012-2015.
6. Thinking in Parallel: Hardware to Software - Adopting the TCPP Core Curriculum in Computer Systems Principles, National Science Foundation TCPP Early Adopter Status, $1000, Spring 2012.
7. Accelerating JavaScript with Safety and Reliability, Trinity College Student Research Program, $3500, Summer 2011.
8. Describing the Operating System for Accurate User-mode Simulation, National Science Foundation CSR SGER, $186,770, 2008-2010.
9. CoGenT: CoGenerating Tools for Modeling Next Generation Systems, National Science Foundation CSR SMA, $299,999, 2006-2009.
10. Bridging the Compiler-Simulator Gap: Co-Optimizing Hardware and Software for Dynamically Compiled Languages, National Science Foundation, $300,000, 2003-2006.
11. Supporting Compiler/Simulator Co-Evolution for Architectural Exploration and Evaluation, National Science Foundation, $49,990, 2002.

Committees and Service

1. Teaching Development Committee, 2022
2. Lecturer Search Committee, 2022
3. Chair of the Curriculum Revision Committee, 2021 – 2022
4. Undergraduate Program Director (UPD), UMass Amherst, 2016 – 2021
5. CARE Undergraduate Teaching, UMass Amherst, 2020 – 2021
6. Teaching Faculty Mentor, 2018 – 2022
7. Senior Celebration Committee, UMass Amherst, 2016 – present
8. Teaching Fellow Search Committee, UMass Amherst, 2017 – 2018
9. Lecturer Search Committee, UMass Amherst, 2015 – 2019
10. Teaching Track Committee, UMass Amherst, 2014 – 2019
11. Scalability Committee, UMass 2014 – 2015
12. Informatics Committee, UMass Amherst, 2013 – 2014
13. Sophomore Advisor, UMass Amherst, 2014 – 2015
14. Chief Undergraduate Advisor (900 CS Students), UMass Amherst, 2012 – 2016
15. Honors Program Director of the Computer Science Honors Program, UMass Amherst, 2011-2012
16. Informatics Committee, UMass Amherst, 2011 – 2012
17. Undergraduate Program Committee, UMass Amherst, 2011 - 2022

Supervised Students

1. Eugene Mak, The Efficiency of the Rust Programming Language, Spring 2022-present (Chair Honors)
2. Vineeth Rao, Modern UI with React, Spring 2022 (IS)
3. Abhinav Tripathy, Scalable Systems Curriculum Generation, Fall 2021 - Spring 2022 (IS)
4. Ishaan Khurana, Teaching and Learning About Scalability, Spring 2022 (IS)
5. Gaurang Chaudhary, Wayfair Spring Co-op Experience, Fall 2021 - Spring 2022
6. Kyle Hesemeyer, Memory Allocation in xv6, Fall 2020 - Spring 2021 (Chair Honors)
7. Aidan Greenslade, Teaching Unix Effectively, Spring 2021 (IS)
8. Kyle Hesemeyer, OS: Memory Allocation and How xv6 Demonstrates it, 2020-2021 (Chair Honors)
9. Craig Norton, Innovation and Entrepreneurship at UMass, 2019 – 2020 (Graduate IS)
10. Sai Magapu, Privacy and Security in Cloud Services, Fall 2019 (IS)
11. Clair Gu, Game Development, Fall 2019 (IS)

12. Qishen Li, Game Development, Fall 2019 (IS)
13. Christopher Doan, Browser Application Development, Fall 2019 (IS)
14. Blake Geraci, Distributed Computing, Fall 2019 (IS)
15. Anjali Devakumar, Enlisting Principles from Cognitive Psychology to Design Secure Email Interfaces, 2019 (Honors)
16. Emily Goroza, Ethical User Experience Design in Social Media, 2018-2019 (Chair Honors)
17. Jonathan Bailey, Lecture Viewer/PAOL, 2017-2018 (IS)
18. Liam Rothschild-Shea, Lecture Viewer/PAOL, 2017-2018 (IS)
19. Rajesh Susai, Lecture Viewer/PAOL, 2017-2018 (IS)
20. Ryan Jerue, Lecture Viewer/PAOL, 2017-2018 (IS)
21. Thomas Aurigemma, Lecture Viewer/PAOL, 2017-2018 (IS)
22. William He, Lecture Viewer/PAOL, 2017-2018 (IS)
23. Dylan Bernard Fischler, Lecture Viewer/PAOL, 2016 (IS, RIPPLES)
24. Mary Moser, Lecture Viewer/PAOL, Fall 2016 (IS)
25. Ryan William Stanly, Lecture Viewer/PAOL, 2015 (IS)
26. Steven Jones, Apple/Android UI Usability, Fall 2015 (Chair Honors)
27. Gary White, Lecture Viewer/PAOL, Fall 2015 (RIPPLES)
28. Ryan Stanley, Lecture Viewer/PAOL, Fall 2015 (RIPPLES)
29. Tri Nguyen, Lecture Viewer/PAOL, Fall 2015 (RIPPLES)
30. Timm Allman, Lecture Viewer/PAOL, 2015
31. Ishita Dasgupta, Lecture Viewer/PAOL , 2015
32. Freddy Nguyen, Lecture Viewer/PAOL, Fall 2014-2015 (RIPPLES)
33. Adam Nelsen, Architecture and Language Implementation, 2014-2015
34. Levi Ramsey, Machine Description Languages, Spring 2014 (IS)
35. Karan Gupta, AI Search Techniques, Spring 2014 (IS)
36. Tyler Karuzis, Web Systems, Spring 2014 (IS)
37. Joshua Chudy, Web Systems, Spring 2014 (IS)
38. Daniel Barrington, Web Systems, Spring 2014 (IS)
39. Lakshmi Guptha Rajagopal, Architecture and Language Implementation, 2013-2015
40. Brendan Murphy, Lecture Viewer/PAOL, Fall 2013-2014 (RIPPLES)
41. Raelen Recto, Research in Presentation Production for Learning Electronically Laboratory, Co-Chair Masters in CS, 2013-2014
42. Kun Tu, Architecture and Language Implementation, 2012-2013
43. Tung Pham, Lecture Viewer/PAOL, Fall 2012-2015 (RIPPLES, Chair Honors)
44. Ryan Szeto, Lecture Viewer/PAOL, Fall 2012-2015t (RIPPLES, Chair Honors)
45. Colin Diedrich, Lecture Viewer/PAOL, Fall 2012-2013 (RIPPLES)
46. Simon Zhang, Lecture Viewer/PAOL, Fall 2012-2013 (RIPPLES)
47. Stefan Valentin, Lecture Viewer/PAOL, Fall 2012-2013 (RIPPLES)
48. Kevin Fernades, Architecture and Language Implementation, Fall 2012-2014 (BDIC)
49. Addison Mayberry, Architecture and Language Implementation, 2011-2012
50. Elisabeth Baseman, Architecture and Language Implementation, 2011-2012
51. Derek Anton, Game Design and Implementation, Fall 2012 (IS)
52. Jordan Moore, Game Design and Implementation, Fall 2012 (IS)
53. Jacob Eid, Programming the Browser, Spring 2012 (IS)